

Retail Sales Analysis Project

Comprehensive analysis of 19M sales transactions revealing strategic insights to optimize marketing, enhance customer engagement, and drive revenue growth across product categories and sales channels.





Project Scope

Understanding Retail Performance

Business Challenge

A leading retail company needed to understand evolving sales patterns to enhance performance, customer satisfaction, and long-term loyalty across demographics, product categories, and sales channels.

Dataset Scale

- **19M records** of sales transactions
- Time frame: 2017-2019
- 19 attributes analyzed
- 37 null records cleaned

Technical Methodology

01

Data Preparation

Python-based EDA with feature engineering, creating temporal metrics and performance indicators from raw transaction data.

03

Advanced Analytics

SQL queries with window functions and CTEs to extract actionable insights from complex patterns.

02

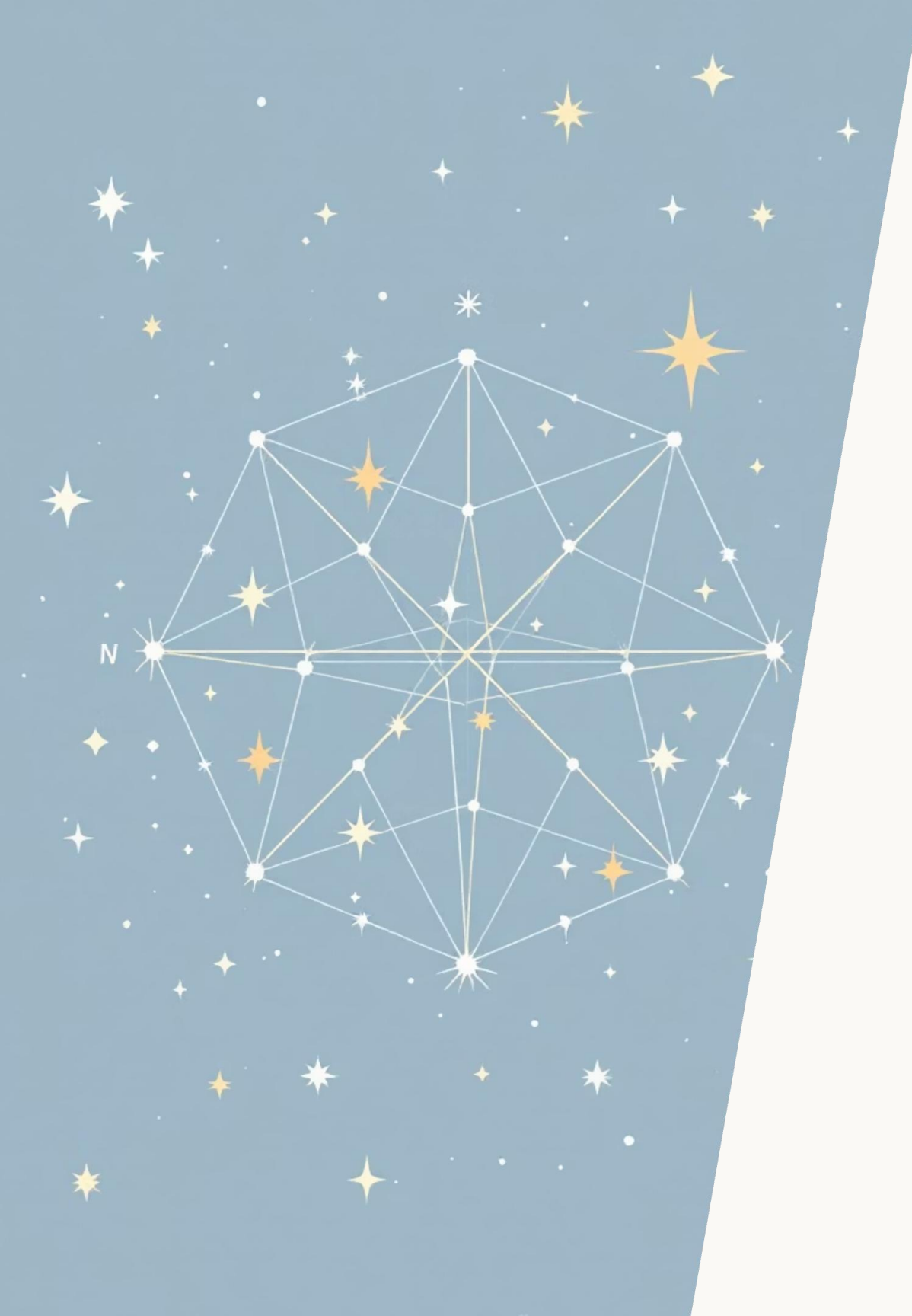
Database Architecture

Star schema implementation in MS SQL Server with fact and dimension tables for optimized querying.

04

Visualization

Power BI dashboards with interactive filtering, drill-through capabilities, and executive reporting.



Data Engineering Excellence

Feature Engineering

- Revenue per Unit calculated
- Stock Turnover metrics
- Temporal features extracted

Star Schema Design

- FactSales: Transaction metrics
- DimProduct: Product hierarchy
- DimStore: Store attributes
- DimDate: Time intelligence

Quality Validation

- Zero missing foreign keys
- Logical value imputation
- Type consistency verified

Product Performance Insights

84%

Revenue Concentration

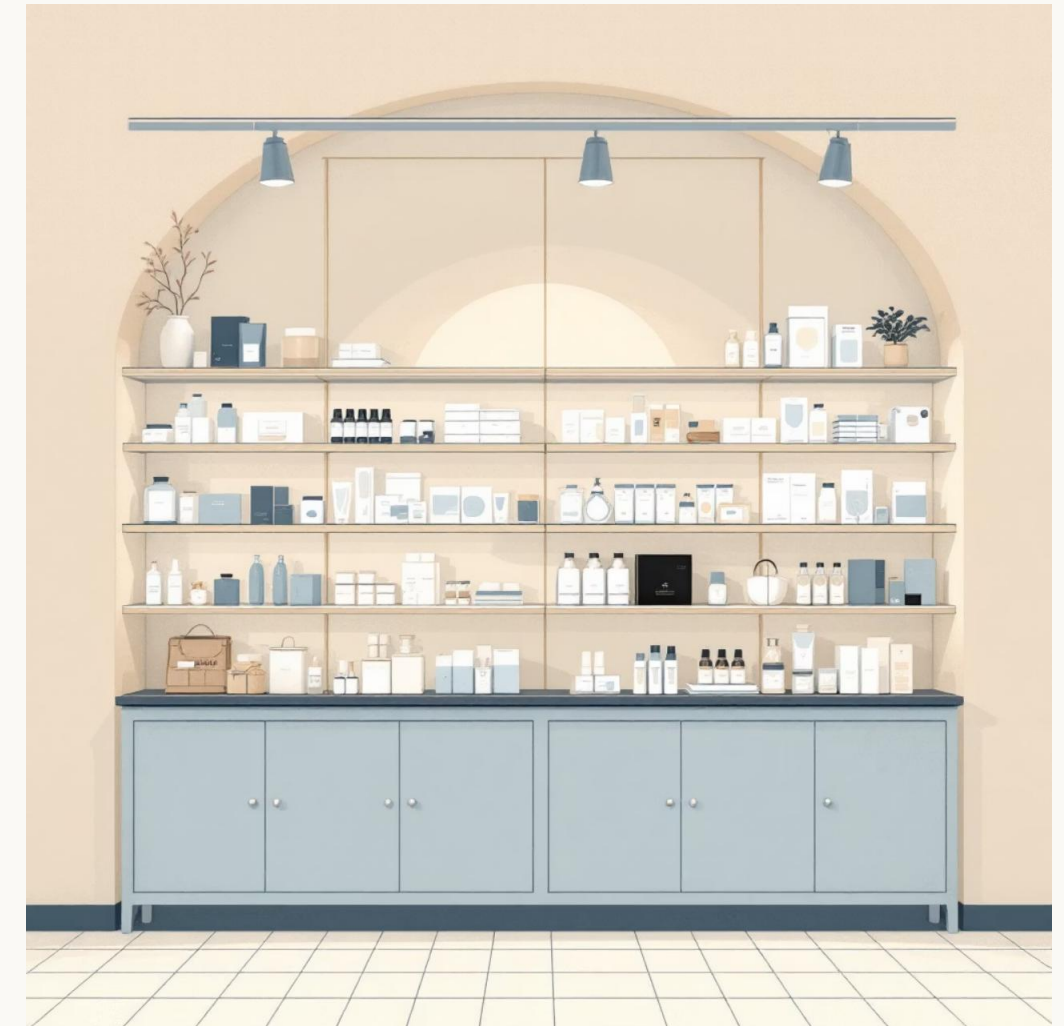
Top hierarchies H00 and H01
generate 84% of total profit

Analysis of hierarchy3 level products revealed significant revenue concentration, enabling targeted inventory and marketing strategies for high-performing categories.

30%

Profit Margin

Assumed margin used for
profitability analysis





Store Performance Analysis

Top Performers

Store types ST04 and ST01 generate **83% of revenue**. These locations demonstrate optimal performance and should serve as models for improvement initiatives.

Space Efficiency

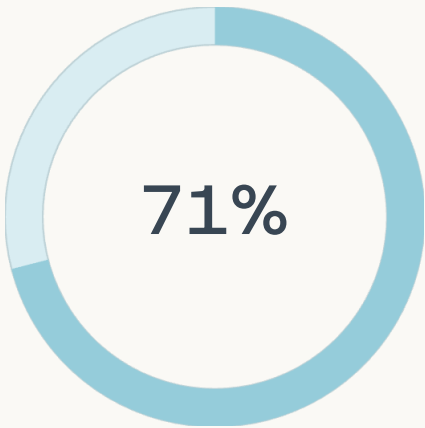
ST04 and ST01 achieve **highest revenue per square meter**, indicating superior space utilization and customer traffic management.

Performance Gap

Bottom 5 stores require **training plans and operational support**. Top performers need recognition programs to maintain excellence.

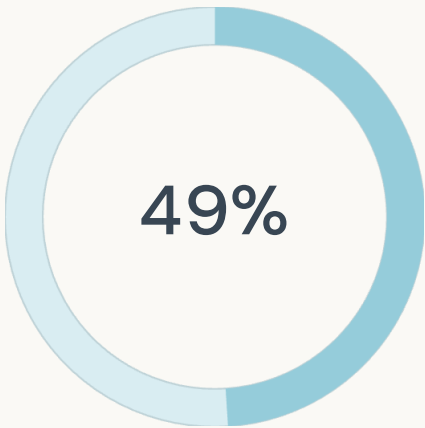
Promotion Campaign Success

Promo Type 1



Sales Impact

PR03 and PR07 combined



PR03 Share

Of promo type 1 sales

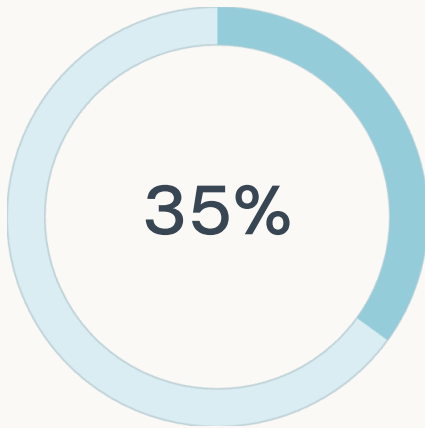
471K total sales achieved through promo type 1 campaigns

Promo Type 2



Dominance

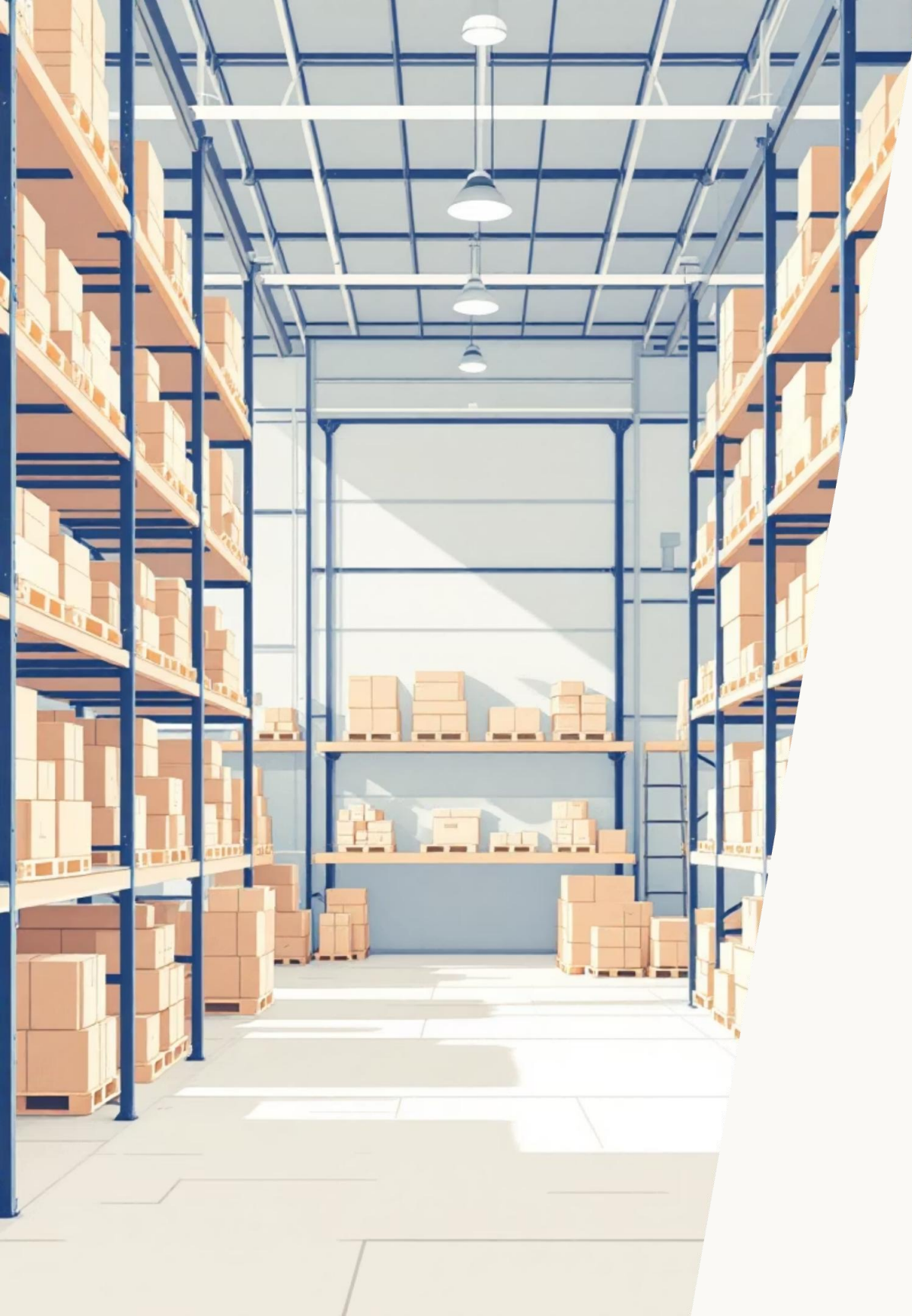
PR03 market share



City Focus

C014 and C022 contribution

19M total sales driven by promo type 2, with PR03 proving exceptionally effective in cities C014 and C022



Inventory & Seasonal Trends



Peak Season

May to August represents highest revenue period, requiring increased inventory and staffing



Stock Turnover

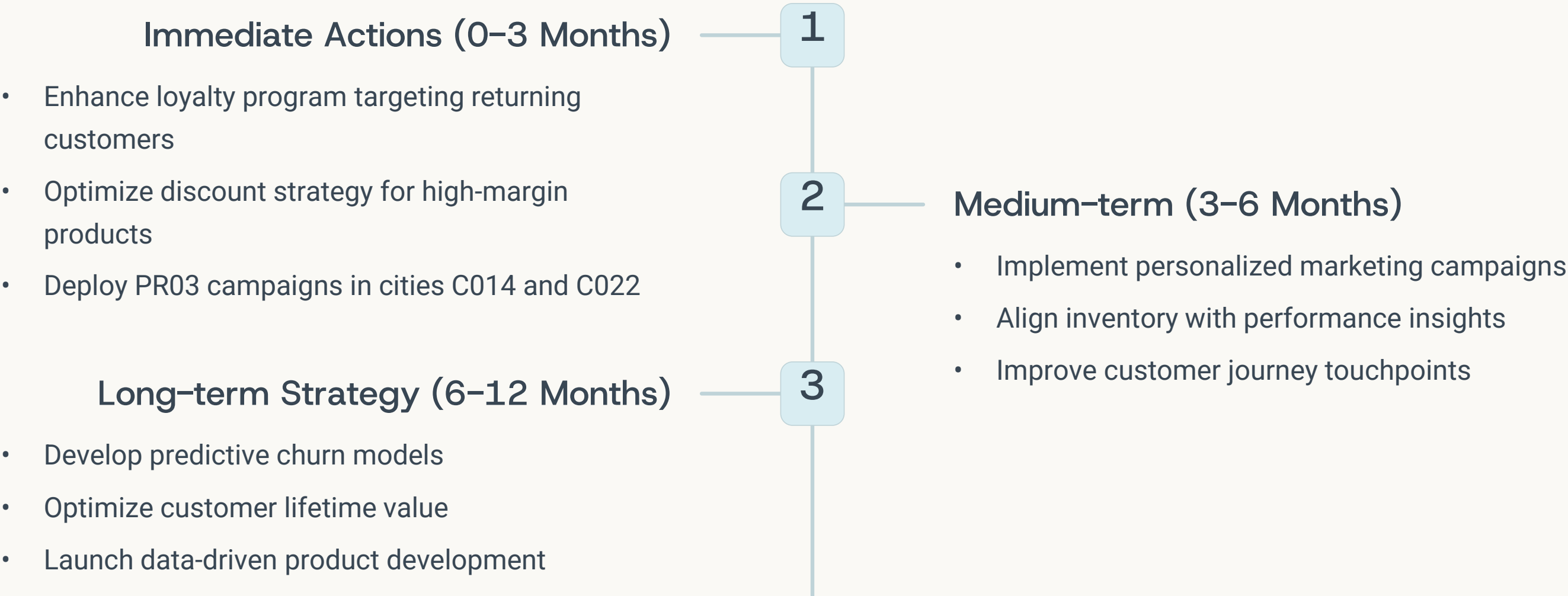
Clusters **3, 7, and 9** represent 45% of turnover; H02 shows 40% of top sales-to-stock ratio



Optimization

Align inventory levels with category performance to reduce holding costs and stockouts

Strategic Roadmap



Impact & Next Steps



Retention

Increase customer loyalty through targeted programs based on segment insights



Optimization

Optimize marketing spend with data-driven, segment-specific strategies



Innovation

Enhance product offerings using performance analytics and customer feedback



Efficiency

Improve operations through pattern recognition and predictive analytics

This scalable technical foundation enables ongoing business intelligence and continuous optimization of retail performance.

